A Switched Optimal Control Approach to Reduce Transferring Time, Energy Consumption, and Residual Vibration of Payload's Skew Rotation in Crane Systems ⋆

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Abstract

In this paper, nonlinear optimal control schemes are proposed to reduce transferring time, energy consumption, and residual vibration for the payload's skew rotation process of crane systems. The main contribution of this paper is to reduce the energy consumption without trading-off the sub-optimal transferring time. The novel idea is to use an electro-mechanical clutch to intelligently disengage the connection between the motor and the payload during the motion such that the payload can continue rotating only by its own momentum. As a result, a switched optimal control problem must be realized. Two solutions, namely particular and general schemes are proposed in the paper, where physical constraints of the actuator including bounded velocity and bounded acceleration are explicitly taken into account. Both simulation and experimental results are provided to prove the effectiveness of the proposed optimal control systems. The established schemes can be directly applied to transfer the payload to a desirable skew orientation without any residual oscillation, or can be utilized as a sub-optimal-time reference trajectory planner of the skewing control module in either overhead or rotary crane systems. Furthermore, the hybrid rotation process presented in this paper, which is driven by the engaging/disengaging event of the clutch, can be served as a theoretical benchmark for any newly established switched optimal control method.

Keywords: Crane Control, Vibration Control, Applied Optimal Control, Switched System, Electro-mechanical Clutch.

1. Introduction

Skew orientation control of a payload plays an important role in crane systems. For instance, consider the overhead crane system shown in Fig. 1 where the payload's center of gravity G has arrived at the correct *position* in the workspace without any oscillation. However, when the target *orientation* is taken into account, the transferring process is said to be incomplete since there exists a skewing mismatch angle β between the payload's orientation and its desirable pose. Therefore, it is essential to rotate the payload at a skew angle of β along the Z axis in the counter-clockwise direction to accomplish the transferring task. A similar issue is also encountered in boom crane systems in use at harbors where the skew orientation of the container must be adjusted to comply with that of the truck or vessel [1].

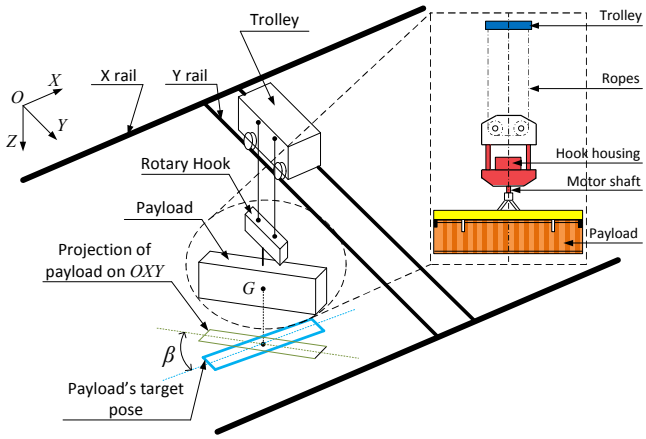

Figure 1: Necessity of the payload orientation control in an overhead crane system. To adjust the payload's skew orientation, the direct mechanism with an aid of the rotary hook device is employed.

In practice, there are two types of mechanisms currently being used to control the skew orientation of the payload, namely the indirect and direct types. The indirect mechanism is usually utilized in a container crane system in which a sophisticated four-rope configuration in conjunction with the usage of a spreader is presented [2, 3, 4]. Four actuators are equipped to independently control the length

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of each rope therefore it is able to control full orientations of the payload, namely trim-list-skew angles. However, in this type of mechanism, the maximum skewing angle that can be adjusted is only up to a few degrees in both clockwise/counter-clockwise directions. Therefore, flexibility of the skew adjustment is very limited. The second kind of skewing control mechanism is illustrated in Fig. 1, where a two-rope configuration in combination with a simple device called a rotary hook are employed. In contrast to the indirect type, the rotary hook device is not designed to control the rope lengths. Alternatively, it will directly control the skewing orientation of the payload by a servo motor which is mounted inside the device. As shown in the zooming area in the Fig. 1, the payload is directly attached to the motor shaft and the entire suspended load is hung below the trolley through two flexible ropes. By using this kind of direct mechanism, the skewing orientation can be controlled to any desired value (> 360 degrees). However, along with the above advantage, after reaching the desired angle, the payload will continue to oscillate rather than stop at the designated skew angle due to the law between inertia force and rope tensions. In practice, the payload inertia is large and the damping ratio of the system is low hence an artificial oscillation suppression needs to be imposed. At present, the payload orientation control through the rotary hook device is manually handled by a highly experienced and skilled operator because of the under-actuated characteristic of the discussed system and as a result, the transfer process is not particularly time efficient. Therefore, an automatic controller should be developed to reduce the workload for the crane operator as well as drastically enhance system productivity. Specifically, two control objectives need to be accomplished simultaneously, that is to say driving the payload to the desired skew angle and suppressing the residual skew oscillation caused by flexible ropes. Especially, this paper will focus on the direct mechanism type with the aid of a rotary hook device to obtain full flexibility in controlling the payload's skew angle.

2. Literature review and contribution of the paper

Despite the fact that numerous anti-swing controllers have been proposed for the crane systems [5, 6, 7, 8, 9, 10, 11, 12], there are only a few researches paying attention to designing automatic controllers for suppressing the residual skew oscillation of the payload. In the context where an overhead crane system is considered, Sawodny et al. [13] introduced a linear model of a rotating system whose function is similar to the rotary hook device. Two degree of freedoms controller, namely feedback in combination with feedforward, is designed based on the pole placement technique. Experimental results show that tracking error and vibration damping capacity of the proposed controller are satisfactory. Nevertheless, a nonlinear model needs to be further developed and employed to enhance control performance as well as feasible operating region. Terashima et al. [14] established an innovative autonomous 3D crane system consisted of a path planning and an obstacle avoidance algorithms considering the rotary motion of the payload. This study only emphasizes the potential method and its extension to solve the path generation problem. In other words, suppression of the skew oscillation is not considered to an elaborate degree. In circumstances with regard to the boom crane system at harbor, Gustafsson [15] formulated a nonlinear model of the rotary hook system in conjunction with the development of two controller types, namely time-optimal and state feedback control based on a simplified double integrator plant. Full-scaled experiments are conducted to verify performance of the state feedback controller where controller gains are chosen based on trial and error. Schaper et al. [16] proposed a 2-DOF skew control with a novel trajectory controller which consists of both state feedback and model predictive control feedforward paths. Performance of the control system is validated with test drives on a full-size mobile harbor boom crane. To improve robust performance of the control system, an Adaptive Input Shaping (AIS) controller is formulated in [17]. The non-conventional algebraic estimator is central to the AIS scheme which allows the Zero Vibration (ZV) input shaper robust to any variation range of natural frequency. Simulation studies are provided to prove the effectiveness of the AIS algorithm. The main drawback of the above researches lies in the fact that only linear control methods are employed. In [1], a nonlinear Sliding Mode Control (SMC) with a coupled linear sliding surface is proposed. Stability of the closed-loop system is proved based on the indirect Lyapunov method to obtain algebraic inequality constraints of the SMC gains. The above study is further developed in [18, 19] where a coupled integral sliding surface is employed as a replacement for the conventional linear type. Certain choices of controller gains are adopted to guarantee the system slide on the sliding surface at the beginning of time as well as to be asymptotically stable in sense of the closed-loop dynamic. Effectiveness of SMCs in [1, 18, 19] are demonstrated on a miniature rotary hook system, in which the results are satisfactory without any gain tuning activities. However, in the aforementioned studies, a sub-optimal-time transferring process (subject to the physical constraints of a real rotary hook system) is not considered, hence productivity is not optimum. Furthermore, the feedback controllers always require a sophisticated and expensive sensor system. For those reasons, in this paper, feedforward control schemes based on the nonlinear optimal control technique will be formulated to obtain a low-cost control system while maintaining a precise skew transferring and an effective residual oscillation suppression. As the main contribution of the paper, the problem in saving energy without sacrificing the sub-optimal transferring time will be elaborately considered by employing the switched optimal control approach with the aid of an electro-mechanical clutch.

In the literature, a clutch is usually used in vehicle trans-

mission control [20, 21, 22, 23]. The main functionality of the clutch in a powertrain system is to allow the car to both change speed and to completely stop without turning off the engine. However, in contrast to the existing researches, this paper will introduce a new application aspect of the electro-mechanical clutch, which is installed inside the rotary hook device, in reducing energy consumption for the skew transferring process of crane systems.

In this paper, indoor application for the overhead crane system is adopted as the context in which the rotary hook system is employed hence external disturbances can be neglected. Simple operations of the clutch, namely engaging and disengaging, will result in a hybrid system dynamic which can be obtained by the Euler-Lagrange approach. The key idea is to disengage the clutch and turn off the motor simultaneously in a specific interval during the transferring process such that the payload can travel only by its own inertial movement. In order to find appropriate switching instants at which the clutch is disengaged or engaged as well as the optimal control input during the actuated intervals, a switched optimal control approach is utilized. To solve the problem, both particular and general solutions are proposed. A particular solution is established by imposing specific state constraints at the switching instants so that the original problem can be transformed to the no-switched domain. On the other hand, the general solution views the switched optimal control problem in its original perspective, namely both switching instants and control inputs need to be optimized. In comparison with the general solution, the particular solution provides better control performance and computational efficiency. By employing the switched optimal control system, the energy consumptions are reduced up to 25.49% and 61.70% in the medium and long transferring cases respectively, *without any influence on the sub-optimal transferring time.*

The remainders of the paper are organized as follows. section 3 introduces the internal structure of the rotary hook device with an electro-mechanical clutch as well as the experimental apparatus. In section 4, modeling of the hybrid rotary hook system is established in theory and validated in experiments. A brief review of the clipping-off Conjugate Gradient (CG) method is given in section 5. Section 6 formulates an optimal control scheme for the no-switched system in which the clutch is engaged during the entire transferring process. Switched optimal controls are then presented in section 7 with both particular and general solutions. Section 8 provides simulation and experimental results of the switched optimal control system. Discussions and comparisons between switched and no-switched scenarios in terms of transferring time and energy consumption are given in section 9. Moreover, input shaping and integral sliding mode controls are applied to highlight advantages of the switched optimal control system. Finally, some conclusions and future works are presented in section 10.

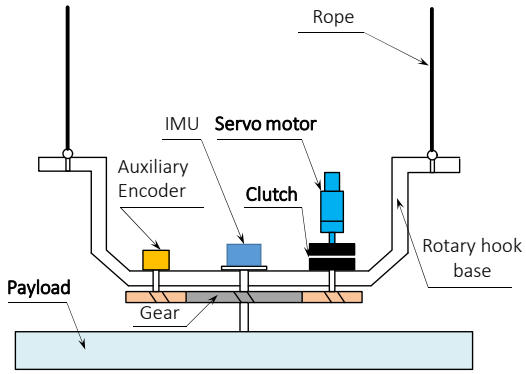

Figure 2: Internal structure of the rotary hook system with a clutch.

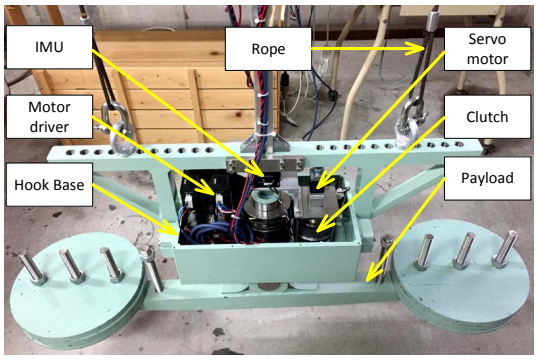

Figure 3: Experimental testbed configuration. A high accuracy 9-DOF IMU is utilized to measure the skew oscillation angle $\theta(t)$. The relative angle $\varphi(t)$ are measured by means of the encoders.

3. Structure of the switched rotary hook system with an electro-mechanical clutch

Fig. 2 shows the internal structure of the rotary hook system with an electro-mechanical clutch. The clutch is comprised of two parts. The rotor (upper part) is connected with the servo motor shaft and the armature (lower part) is attached to a transmission gear system whose end of effector is the payload. When the current is applied, the clutch is engaged which means that the upper and lower portions are connected with each other thus the servo motor can transmit the motion to the payload. On the other hand, removing the current will cause two parts of the clutch to separate hence the payload cannot be actuated by the servo motor, the clutch is said to be in the disengaging state. In this case, the system is constituted of two autonomous subsystems, namely the hook device and the payload. If the initial absolute velocity of the payload in the disengaging period is not identical to zero then the payload will continue rotating relying upon its own momentum, whereas the hook keeps oscillating with a higher natural frequency than in the engaging case due to reduction of the overall moment of inertia. In order to verify the effectiveness of the proposed control schemes, an experimental apparatus is built, whose configuration is illustrated in Fig. 3. The actuator is a TAMAGAWA TS4738

servo motor. The incremental encoder attached at the end of the motor has a resolution of 2048 pulses/rev. For safety reasons as well as according to the motor and gear transmission ratio specifications, the maximum speed of the servo motor is given by $v^m_{\max} = 2000$ rpm $= 209.44$ rad/s and the limitation torque of the servo motor actuating on the payload is $\tau_{\max} = 6.6$ (Nm). Note that when the clutch is disengaged, the position of the payload cannot be measured by the encoder attached at the end of servo motor. Therefore, an auxiliary encoder with a resolution of 2048 pulses/rev is installed to retrieve the rotation information of the payload in the disengaging state.

4. Mathematical modeling validation

4.1. Mathematical modeling

This section is devoted to formulating a modeling of the rotary hook system which consists of a rotary hook device, payload, ropes, and crane tip. The skew residual oscillation resulting from the rotation of the payload is depicted in Fig. 4, where $\varphi(t)$ denotes the relative angle between the payload and the hook, whereas $\theta(t)$ and $\gamma(t)$ represent the absolute angles between the hook and the payload versus x axis respectively. Note that two control objectives, namely driving the payload to a reference skew angle γ_d ($\gamma(t) \to \gamma_d$ and $\dot{\gamma}(t) \to 0$) without any vibration ($\theta(t) \to 0$ and $\dot{\theta}(t) \to 0$) are utilized to establish the cost functionals. Table 1 describes the definitions of all system parameters.

In Fig. 4, P refers to the conjunction point of the load and the hook. Denote the length of OP line as a. The steel ropes are assumed to be inelastic thus following holonomic constraint must hold

$$a^2 = l^2 - 2R^2(1 - \cos\theta). \tag{1}$$

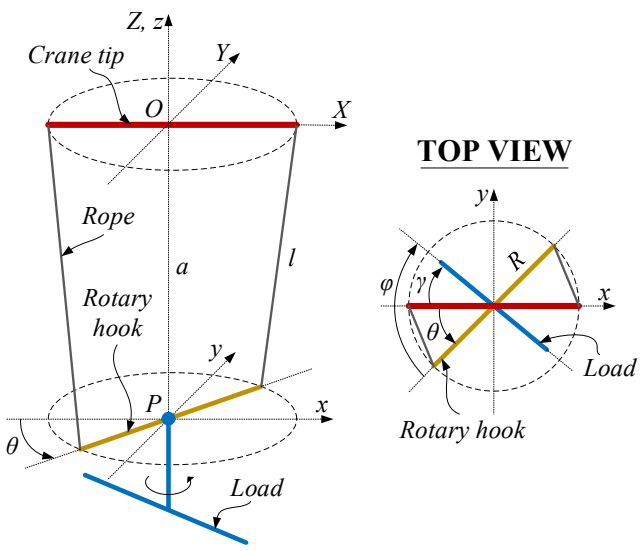

Figure 4: Modeling of the rotary hook system.

Table 1: Parameters of the miniature rotary hook system.

Symbol	Description	Value
I_L	Inertia of the payload (kgm^2)	16.5
I_H	Inertia of the hook (kgm^2)	1.25
m	Total mass of the load and the hook (kg)	142.2
R	Skewing radius (m)	0.25
l	Rope length (m)	4
g	Gravity acceleration (m/s^2)	9.81

The total kinetic energy of the system comprises of the rotation energies of the payload and the hook as well as the translation energy of the overall mass along the vertical direction. Therefore, it can be written by

$$T = \frac{1}{2}I_L\dot{\gamma}^2 + \frac{1}{2}I_H\dot{\theta}^2 + \frac{1}{2}m\dot{a}^2. \tag{2}$$

The potential energy of the system is given as

$$V = -mga. \tag{3}$$

Therefore, the Lagrangian function can be established by

$$L = T - V = \frac{1}{2}I_L\dot{\gamma}^2 + \frac{1}{2}I_H\dot{\theta}^2 + \frac{1}{2}m\dot{a}^2 + mga. \tag{4}$$

The generalized coordinate vector of the system is chosen as $\boldsymbol{q} = \begin{bmatrix}\gamma(t) & \theta(t)\end{bmatrix}^\top$. Applying the Euler-Lagrange equation on the first generalized coordinate $\theta(t)$ yields

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0. \tag{5}$$

Note that $\gamma(t) = \varphi(t) - \theta(t)$. By developing (5) using (1) and (4), it can be obtained that

$$\left(I_H + \frac{mR^4\sin^2\theta}{a^2}\right)\ddot{\theta} = -\frac{mR^2\sin\theta}{a}\left(\frac{R^4\dot{\theta}^2\sin^2\theta}{a^3} + \frac{R^2}{a}\dot{\theta}^2\cos\theta + g\right) + I_L\ddot{\gamma}. \tag{6}$$

Denote $\tau(t)$ as the net torque exerting on the payload. Employing the Euler-Lagrange equation on the actuated generalized coordinate $\gamma(t)$ results in

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\gamma}}\right) - \frac{\partial L}{\partial \gamma} = \tau. \tag{7}$$

By developing (7), the second dynamic equation can be adopted. By (4), following results can be easily obtained

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\gamma}}\right) = I_L\ddot{\gamma}, \quad \frac{\partial L}{\partial \gamma} = 0. \tag{8}$$

Substituting (8) into (7), it can be obtained that

$$I_L\ddot{\gamma} = \tau. \tag{9}$$

It is noted that one can either use the acceleration $\ddot{\gamma}(t)$ or the torque $\tau(t)$ as the control input for the system since they just relate to each other through a constant I_L. However, for the sake of convenience, the acceleration $\ddot{\gamma}(t)$ will be utilized in this paper. Therefore, the system dynamic during the engaging period of the clutch is described by (6) in which the control input is the acceleration $\ddot{\gamma}(t)$ generated from the servo motor.

When the clutch is disengaged, there are two subsystem motions exist, namely the free-running payload and the free-oscillating hook device. In such circumstance, by assuming that the friction is sufficiently small, the torque acting on the payload is zero, namely $\tau = 0$; hence, the free motion of the payload during the disengaging period of the clutch is governed by

$$I_L \ddot{\gamma} = 0. \tag{10}$$

Note that (10) is equivalent to the result of angular momentum conservation principle, and it is only used when the clutch is disengaged. Since the payload inertia I_L does not vary with time; hence, from (10), the payload will travel with a constant velocity. In addition, the free oscillating movement of the rotary hook device can be depicted by substituting (10) into (6).

It is worth mentioning that from (9), by neglecting the friction forces, one can easily compute $u_{\max}$ as $u_{\max} := |\ddot{\gamma}|_{\max} = \tau_{\max}/I_L = 0.4$ (rad/s^2). Furthermore, denote the speed of the servo motor as $\dot{\varphi}_m(t)$ ($|\dot{\varphi}_m(t)| \leq v^m_{\max} = 209.44$ (rad/s)). Based on the transmission ratio of the auxiliary gear system (see Fig. 2), it holds that $\dot{\varphi} = \dot{\varphi}_m/r_g$ ($r_g = 400$) when the clutch is engaged. Therefore $v_{\max} := |\dot{\varphi}|_{\max} = v^m_{\max}/r_g = 0.524$ (rad/s) during the engaging state of the clutch. Note that when the clutch is disengaged, generally $\dot{\varphi} \neq \dot{\varphi}_m$.

In summary, the state of the electro-mechanical clutch, that is to say engaging and disengaging, will determine the dynamic of the switched rotary hook system which falls into either of the following configurations

$$\text{Engaging:} \quad \begin{cases} \ddot{\gamma} = u \\ \ddot{\theta} = f(\theta, \dot{\theta}) + b(\theta)u, \end{cases} \tag{11}$$

$$\text{Disengaging:} \quad \begin{cases} \ddot{\gamma} = 0 \\ \ddot{\theta} = f(\theta, \dot{\theta}), \end{cases} \tag{12}$$

where

$$\begin{aligned} f(\theta, \dot{\theta}) &= \frac{1}{I_H + \dfrac{mR^4 \sin^2\theta}{a^2}} \times \\ &\quad \left[-\frac{mR^2 \sin\theta}{a} \left(\frac{R^4 \dot{\theta}^2 \sin^2\theta}{a^3} + \frac{R^2 \dot{\theta}^2 \cos\theta}{a} + g \right) \right], \\ b(\theta) &= \frac{I_L}{I_H + \dfrac{mR^4 \sin^2\theta}{a^2}}. \end{aligned} \tag{13}$$

Note that the nonlinear system (12) which describes the free motions of the load and the hook are always autonomous since no control input can be applied when the clutch is disengaging.

Remark 1. In this paper, no-switched system refers to the configuration of the rotary hook system in the context where the clutch is engaged in entire transferring process. On the other hand, the switched system corresponds to the case that the clutch is disengaged at least once during the motion.

Remark 2. The only dimension that can be directly controlled by the servo motor is related to the relative angle $\varphi(t)$, namely $\dot{\varphi}(t)$ and $\ddot{\varphi}(t)$, thus the chosen control input $u(t) = \ddot{\gamma}(t)$ must be regarded as a virtual quantity. In the actual experiment, the virtual optimal control input will be replaced by a profile of the payload's relative speed $\dot{\varphi}(t)$ which is obtained in the simulation. In the other words, the feed-forward speed command will be generated by the simulating trajectory of $\dot{\varphi}(t)$. More specifically, once the optimal profile of the virtual control input $\ddot{\gamma}(t)$ is found, it can be plugged into (6) to numerically compute $\dot{\theta}(t)$ by any ODE solver (e.g. 4th Runge-Kutta method). In addition, $\dot{\gamma}(t)$ can be adopted by integrating $\ddot{\gamma}(t)$. Therefore, one can compute the profile of $\dot{\varphi}(t)$ by using the relation $\dot{\varphi}(t) = \dot{\gamma}(t) + \dot{\theta}(t)$. The actual velocity command to the servo motor can be easily obtained by using $\dot{\varphi}_m = r_g \dot{\varphi}$ where $r_g = 400$.

4.2. Modeling validation

In order to verify the correctness of the established mathematical modeling of the hybrid nonlinear rotary hook system, experimental validations are conducted in the following cases

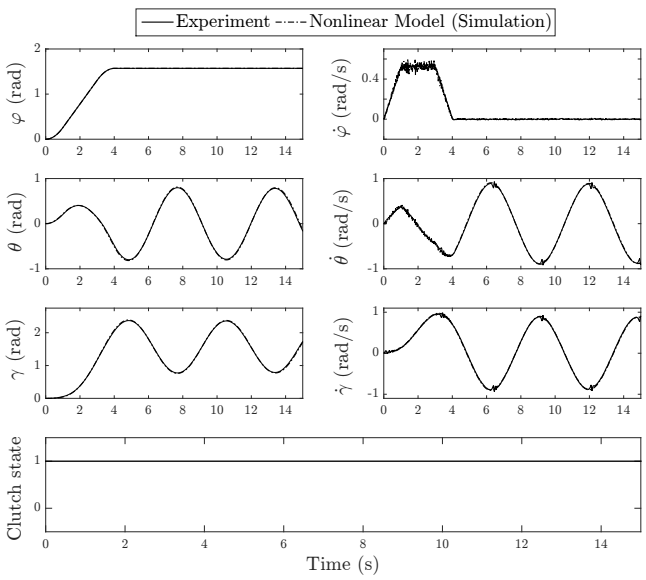

Figure 5: Model validation in the scenario of Case 1: no-switched system.

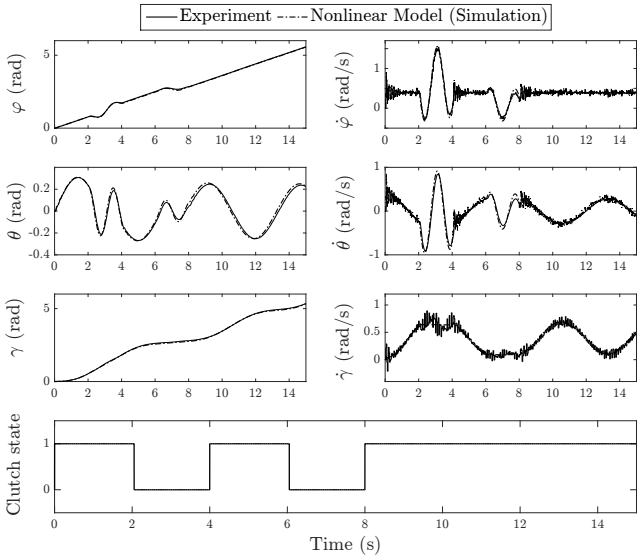

Figure 6: Model validation in the scenario of Case 2: switched system, switching period is 2 seconds.

- Case 1: The clutch is engaged in the entire process, namely the no-switched system dynamic is put into consideration.
- Case 2: During the transferring process, the clutch is engaged and disengaged alternatively at an interval of 2 seconds. By means of this experiment, the established switched system dynamic can be verified.

In all cases, the actual system parameters of the experimental testbed given in Table 1 will be utilized for the nonlinear models (11) and (12) in the simulations. A reference velocity profile of $\dot{\varphi}_m(t)$ is fed to the servo motor and all of system states are measured. Specifically, in Case 1 where the no-switched system is examined, a trapezoidal trajectory of $\dot{\varphi}_m(t)$ is employed. Moreover, in Case 2 where the switched system is under consideration, motion of the servo motor is kept constantly at 1500 rpm. Meanwhile, the clutch is engaged and disengaged alternatively with a period of 2 seconds.

Results of the model validation in two cases are shown in Fig. 5 and Fig. 6 respectively. In each figure, "0" value of the clutch state graph indicates the disengaging interval whereas "1" implies the engaging period of the clutch. Note that the measurement data in Fig. 6 is more fluctuated than Fig. 5 of the no-switched case. This observation can be explained as follows. In Case 2, for the sake of consistency, the payload position information is entirely measured by the auxiliary encoder whereas in Case 1, these data can be obtained by the encoder attached at the end of the servo motor. Furthermore, the noise level of the auxiliary encoder is much heavier than the encoder of the servo motor due to effect of the motor gearbox.

It is recognized that in all cases, experimental data strongly agrees with the mathematical model given in (11) and (12). Therefore it can be concluded that the formulated modeling accurately describes the dynamic behavior of the switched rotary hook system. Note that as shown in Fig. 6, the reduction of the payload velocity $\dot{\gamma}(t)$ during the disengaged stage of the clutch (2 - 4 seconds and 6 - 8 seconds) is very small. It indicates that the friction coefficient of the system can be negligible. In the real application, the internal structure of the hook device is placed inside a closed shield, which protects it from effects of the environment. Therefore, the friction of the system could be guaranteed small without a frequent maintenance.

At this point, the mathematical modeling of the hybrid rotary hook system is successfully established. The optimal control designs for both no-switched and switched systems will be discussed in sections 6 and 7 respectively. In practice, the physical constraints of the actuator must taken into account, thus in this paper, both control input and state constraints will be considered in the optimal control formulations. Generally, it is almost impossible to obtain an analytical solution of the constrained optimal control problem for a nonlinear system except in some simple cases [24, 25, 26]. Therefore, a numerical method will be adopted in this paper.

5. Preliminary of the clipping-off conjugate gradient method

In the domain of numerical approaches to solve a constrained optimal control problem, direct [27, 28, 29] and indirect [30, 31, 32] methods can be found in the literature. However, the indirect techniques always result in a faster convergence rate and better cost performance index [26, 33], thus it will be employed in this paper. Quintana and Davison [32] introduced a novel conjugate gradient method in conjunction with a clipping-off operator to effectively solve the bounded optimal control problem. This technique elegantly incorporates the classical conjugate gradient method (for solving an unconstrained optimization problem) and the clipping-off operator to enforce the local optimizer satisfying the necessary condition of optimality subjected to the control input constraints. For the sake of convenience in following presentations, the algorithm of [32] will be briefly reviewed in this section.

Consider the following class of nonlinear systems

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \quad \boldsymbol{x}(t_0) = \boldsymbol{x_0}, \tag{14}$$

where $\boldsymbol{f}$ is a nonlinear n-dimensional vector function of n-dimensional state vector $\boldsymbol{x}$ and r-dimensional control input $\boldsymbol{u} = [u_1 \ \ldots \ u_r]^\top$ (i.e. $\boldsymbol{f} \in \mathbb{R}^n$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathbb{R}^r$). Denote t_0 and t_f as the initial and terminal time respectively, which are assumed to be known. The optimal control problem with a fixed terminal time t_f is given as

$$\begin{aligned} \min_{\boldsymbol{u}} \quad & J(\boldsymbol{u}) = \phi(\boldsymbol{x}(t_f), t_f) + \int_{t_0}^{t_f} L(\boldsymbol{x}, \boldsymbol{u}) dt \\ \text{s.t.} \quad & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}), \\ & \boldsymbol{u} \in \Omega, \end{aligned} \tag{15}$$

therein the admissible control set Ω is defined as

$$\Omega \triangleq \{u_j : |u_j| \leq u_{\max} \quad (j = 1, 2, \ldots, r)\}. \tag{16}$$

Denote H as the Hamiltonian function

$$H(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{u}) \triangleq \boldsymbol{\lambda}^\top \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) + L(\boldsymbol{x}, \boldsymbol{u}), \tag{17}$$

where the n-dimensional costate vector $\boldsymbol{\lambda}(t)$ must satisfy

$$\dot{\boldsymbol{\lambda}}(t) = -\left(\frac{\partial H}{\partial \boldsymbol{x}}\right)^\top \text{ and } \boldsymbol{\lambda}(t_f) = \left(\frac{\partial \phi(\boldsymbol{x}(t_f), t_f)}{\partial \boldsymbol{x}(t_f)}\right)^\top. \tag{18}$$

In (15), $\phi(\boldsymbol{x}(t_f), t_f)$ is the component related to the terminal condition, whereas $L(\boldsymbol{x}, \boldsymbol{u})$ is corresponding to the accumulating cost performance of the system as the time proceeding. Note that (15) is a general optimal control problem, which is only used for a purpose of briefly reviewing the CG gradient method. Concrete optimal control problems for the no-switched and switched rotary hook system are posed in (20), (26), and (29).

Let $\boldsymbol{g}(t) = (\partial H/\partial \boldsymbol{u})^\top = [\partial H/\partial u_1 \quad \ldots \quad \partial H/\partial u_r]^\top \triangleq [g_1(t) \quad \ldots \quad g_r(t)]^\top$, following control input constraints must be fulfilled in order to guarantee the necessary condition for optimal solution [34]

$$\begin{cases} u_j = u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} < 0 \\ -u_{\max} \leq u_j \leq u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} = 0 \\ u_j = -u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} > 0 \end{cases}, \quad j = 1, \ldots, r. \tag{19}$$

Basically, the clipping-off CG method of [32] iteratively compute the control input $\boldsymbol{u}$ from an initial guess until a local optimum solution is reached. Moreover, the resultant local minimizer of (15) will satisfy all of the necessary conditions for optimality given by in (18) and (19). Further details are greatly explained in [32]. It is highly recommended for interest readers.

6. Optimal control of no-switched system

In this section, an optimal control scheme will be formulated for the rotary hook system when the clutch is always kept engaged during the transferring process. Two control objectives, namely driving the payload to the desired skew angle and suppressing the skew oscillation simultaneously (i.e. forcing $\gamma(t) \to \gamma_d$, $\dot{\gamma}(t) \to 0$, $\theta(t) \to 0$, and $\dot{\theta} \to 0$) are the basics to establish the cost functional. A fixed terminal time optimal control problem is firstly formulated and then a sub-optimal-time solution can be obtained by gradually reducing the terminal time through a bisection method.

6.1. Problem formulation and algorithm

Denote a state vector of the control system as $\boldsymbol{x} = [x_1 \quad x_2 \quad x_3 \quad x_4]^\top = [\gamma(t) \quad \dot{\gamma}(t) \quad \theta(t) \quad \dot{\theta}(t)]^\top$. By taking the physical limitations of the servo motor including the maximum velocity $v_{\max}$ and maximum acceleration $u_{\max}$ into account, the optimal control problem for the no-switched rotary hook system can be posed as

$$\begin{aligned} \min_u \; & J = \omega_1 (x_1(t_f) - \gamma_d)^2 + \omega_2 x_2^2(t_f) + \omega_3 x_3^2(t_f) + \omega_4 x_4^2(t_f) \\ s.t. \; & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{b}(\boldsymbol{x})u, \\ & (x_2 + x_4)^2 \leq v_{\max}^2, \\ & |u| \leq u_{\max}. \end{aligned} \tag{20}$$

In (20), t_f is the fixed terminal time and the weighting coefficients are chosen as $\omega_1 = \ldots = \omega_4 = 10^6$. As discussed in section 3, $u_{\max} = |\ddot{\gamma}|_{\max} = 0.4$ (rad/s^2) and $v_{\max} = |\dot{\varphi}|_{\max} = |x_2 + x_4|_{\max} = 0.524$ (rad/s). The system modeling in the engaging state of the clutch given by (11) is employed, thus

$$\boldsymbol{f}(\boldsymbol{x}) = [x_2 \quad 0 \quad x_4 \quad f]^\top \text{ and } \boldsymbol{b}(\boldsymbol{x}) = [0 \quad 1 \quad 0 \quad b]^\top, \tag{21}$$

where f and b terms are given in (13). Note that the optimal control problem (20) is involved by both control input and state constraints. As discussed in section 5, control input constraint can be efficiently solved by the clipping-off CG method. To eliminate the state constraint, the original cost functional J will be augmented by a step penalty function. The state-constraint-free optimal control problem of the no-switched rotary hook system can be further described as

$$\begin{aligned} \min_u \quad & J_s = J + \underbrace{\int_{t_0}^{t_f} k_1 \phi_1^2 dt}_{\text{Penalty term}} \\ s.t. \quad & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{b}(\boldsymbol{x})u \\ & |u| \leq u_{\max} \\ & \phi_1 = v_{\max}^2 - (x_2 + x_4)^2 \\ & k_1 = \begin{cases} 0 & \text{if } \phi_1 \geq 0 \\ \mu_1 & \text{if } \phi_1 < 0, \end{cases} \end{aligned} \tag{22}$$

where $\mu_1 = 10^6$. Note that the penalty coefficient k_1 is activated whenever the state constraint represented in ϕ_1 is violated.

Remark 3. It is easy to reduce the energy consumption of the transferring process by adding the control input to the integral term of (22). However, it is well-known that such treatment will significantly lengthen the transferring time. In the other words, an unfavorable trade-off will be encountered. Alternatively, section 7 will propose a method to save energy of the overall process without influencing the sub-optimal transferring time by employing the switched optimal control approach.

Denote $u^\star(t_f)$ as the resultant optimal control input of (22) with respect to the fixed terminal time t_f. The sub-optimal-time control problem can be solved by extending (22) through a bisection method [35, 36] by means of Algorithm 1. Note that $u^\star(t_f)$ in Algorithm 1 is always obtained by the clipping-off CG method shown in section 5.

Algorithm 1: Bisection method to solve the sub-optimal time problem.

Step 1 Let $t_f = \dfrac{\gamma_d}{\alpha v_{\max}}$, $a_1 \leftarrow 0$, and $a_2 \leftarrow t_f$. Obtaining the optimal control input $u^\star(t_f)$. If $J_s(u^\star, t_f) < J_{tol}$ then to step 2, otherwise go to step 3.

Step 2 Let $t_f \leftarrow (a_1 + a_2)/2$. Retrieving $u^\star(t_f)$. If $J(u^\star, t_f) < J_{tol}$ then $a_2 \leftarrow t_f$, otherwise $a_1 \leftarrow t_f$, go to Step 4.

Step 3 Let $a_1 \leftarrow a_2$, $a_2 \leftarrow 2a_2$, and $t_f \leftarrow a_2$. Calculating $u^\star(t_f)$. If $J_s(u^\star, t_f) > J_{tol}$ then repeat Step 3, otherwise go to Step 2.

Step 4 If $a_2 - a_1 > \Delta_t$ then go to Step 2. Otherwise, go to Step 5.

Step 5 If $J_s(t_f = a_1) < J_{tol}$ then $t_f \leftarrow a_1$, otherwise $t_f \leftarrow a_2$. Output $u^\star(t_f)$ as the optimal control input corresponding to the sub-optimal transferring time t_f.

In Algorithm 1, α relates to the initial value of the terminal time t_f. Typically, it can be chosen that $\alpha = 0.5$. Moreover, J_{tol} specifies the requirements corresponding to the transfer accuracy, residual vibration suppression, and state constraint assurance. With the large weighting coefficients ($\omega_1 = \ldots = \omega_4 = 10^6$), it can be heuristically chosen that $J_{tol} = 1$ to sufficiently fulfill the control performance. The terminated condition of Algorithm 1 is determined by the tolerance Δ_t which regards to the minimum distance between two consecutive trial terminal time points. Here, $\Delta_t = 0.1$ second is adopted. It is important to remark that the initial guess of the optimal control input for the present terminal time t_f should be taken from that of the previous successive iteration. Such choice can significantly reduce the computational effort.

6.2. Simulation and experimental results

By employing Algorithm 1, optimal control with both input and state constraints will be computed for three cases: short ($\gamma_d = 30$ degrees), medium ($\gamma_d = 90$ degrees), and long transferring case ($\gamma_d = 180$ degrees). An initial guess of the optimal control input in the clipping-off CG algorithm is given by $u(t) = 0, t_0 \leq t \leq t_f$. The computation is conducted on a personal computer with 2.5-GHz Intel Core i5-3210M and 8 GB of RAM. The computational results for all cases are summarized in Table 2. Note that the clipping-off CG method is restarted every 2 iterations.

Table 2: Computation results of the no-switched optimal controls.

Case	$\min(t_f)$	J_s	Compt. time	$\max(-\phi_1(t))$
$\gamma_d = 30°$	3.13 s	0.6651	35.1 s	0.0170
$\gamma_d = 90°$	5.06 s	0.8060	61.7 s	0.0093
$\gamma_d = 180°$	7.97 s	0.4165	85.2 s	0.0072

Compt. time: Computational time.

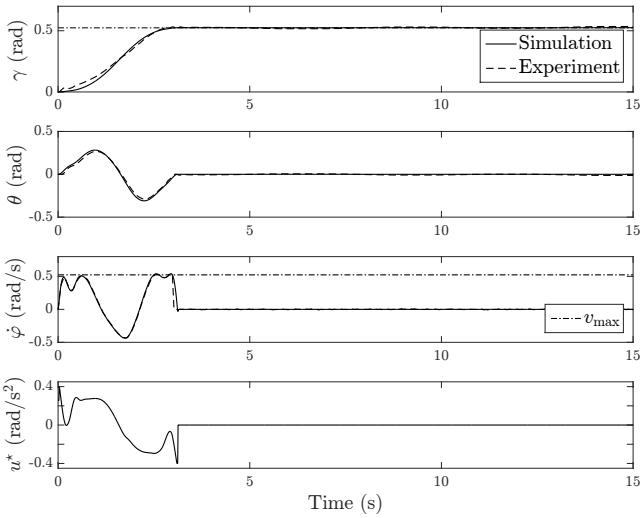

Figure 7: Simulation and experimental results in the short transferring case ($\gamma_d = 30$ degs): no-switched scenario.

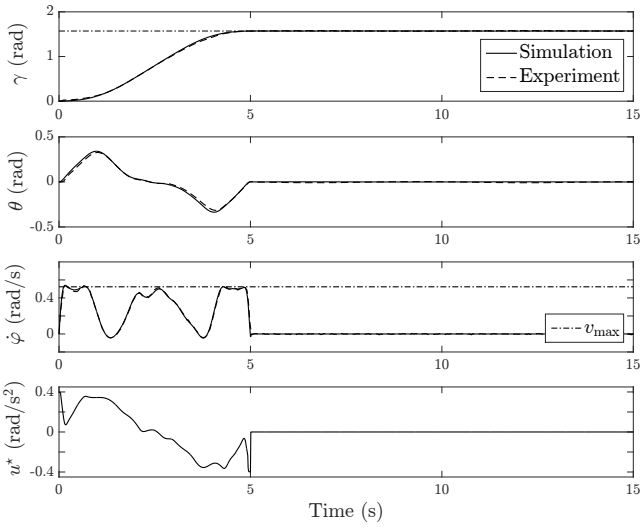

Figure 8: Simulation and experimental results in the medium transferring case ($\gamma_d = 90$ degs): no-switched scenario.

The simulation and experimental results in three cases are shown in Fig. 7 - Fig. 9 respectively. It can be recognized that with the computed optimal control inputs, the payload accurately reaches the desired skew angle as well as the skew oscillation nearly vanishes after the motion. Both control input and state constraints are sufficiently assured by the resultant optimal solutions. Therefore, it is concluded that effectiveness of the proposed no-switched optimal control system is verified.

7. Optimal control of switched system

7.1. Motivation and problem formulation

Consider the motor speed profile $\dot{\varphi}(t)$ of the long transferring case as shown in Fig. 9, one can recognize that the motor speed residents on its upper constraint in the middle

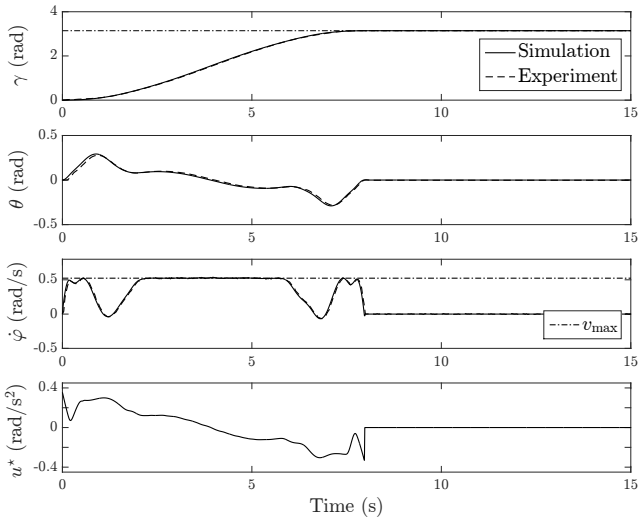

Figure 9: Simulation and experimental results in the long transferring case ($\gamma_d = 180$ degs): no-switched scenario.

of transferring process. The motor must maintain a constant maximum speed during this uniform interval, thus the energy is considerably consumed. Therefore, the basic idea is to remove the above-mentioned undesirable interval by imposing upon the movement generated by the inertia of the payload itself. Specifically, consider the free motion of the payload subsystem when the clutch is disengaged as in (12), namely $\ddot{\gamma}(t) = 0$ whose solution is $\dot{\gamma}(t) =$ constant. It is recognized that the motion of payload in this case has the same property with the unfavorable interval encountered in the no-switched optimal control solution. In the other words, motion of the motor should be replaced by the inertial movement of the payload by disengaging the clutch in a specified period of the entire transferring process to reduce the energy consumption.

By the above observation, the whole time span $\mathbb{T} = \{t : t \in [t_0 \quad t_f]\}$ of the payload motion can be divided into three sub-intervals: $\mathbb{T}_1 = \{t : t \in [t_0 \quad t_1]\}$, $\mathbb{T}_2 = \{t : t \in [t_1 \quad t_2]\}$, and $\mathbb{T}_3 = \{t : t \in [t_2 \quad t_f]\}$. In the first interval $\mathbb{T}_1$, the control input generated by the motor is applied to accelerate the payload until the time variable t reaches the time instant t_1. At t_1, the clutch is disengaged in the second interval $\mathbb{T}_2$ to force the payload travel with a constant speed only by its own inertia. Finally, in the third interval $\mathbb{T}_3$, the clutch is engaged again, the control input is put into operation to complete the entire motion. By means of the above discussion, subsystems and their activated intervals can be described as

$$\begin{array}{lll} \text{Subsystem 1:} & \dot{\boldsymbol{x}} = \boldsymbol{f}_1(\boldsymbol{x}) + \boldsymbol{b}_1(\boldsymbol{x})u_1 & , t \in \mathbb{T}_1. \\ \text{Subsystem 2:} & \dot{\boldsymbol{x}} = \boldsymbol{f}_2(\boldsymbol{x}) & , t \in \mathbb{T}_2. \\ \text{Subsystem 3:} & \dot{\boldsymbol{x}} = \boldsymbol{f}_3(\boldsymbol{x}) + \boldsymbol{b}_3(\boldsymbol{x})u_3 & , t \in \mathbb{T}_3. \end{array} \tag{23}$$

In (23), $\boldsymbol{f}_1(\boldsymbol{x}) = \boldsymbol{f}_2(\boldsymbol{x}) = \boldsymbol{f}_3(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x})$ and $\boldsymbol{b}_1(\boldsymbol{x}) = \boldsymbol{b}_3(\boldsymbol{x}) = \boldsymbol{b}(\boldsymbol{x})$, where $\boldsymbol{f}(\boldsymbol{x})$ and $\boldsymbol{b}(\boldsymbol{x})$ are given in (21). Note that the subsystem 2 is an autonomous system without control input.

Here, there are two switching instants, namely t_1 and t_2 hence it can be termed as a 2-switched control system. Obviously, one can generate N-switched system by engaging and disengaging the clutch N times in which $N = 2k$ ($k \in \mathbb{N}^+$) since at the end of the transferring process, the clutch must be in the engaging state. However, a large number of switchings within a short time possibly cause damage to the clutch. Therefore, a 2-switched control system is reasonably adopted in practice. Note that, the speed profile $\dot{\varphi}(t)$ of Fig. 9 obtained in the no-switched scenario is only employed for a purpose of illustrating the underlying motivation. In the other words, the switching instants as well as the optimal control input for the switched system will be computed in light of the particular solution (section 7.2) or the general solution (section 7.3).

Remark 4. For a mechanical system, physically, it is not feasible to reduce both maneuver time and energy consumption simultaneously. Therefore, the best circumstance can be expected is that, by using the switched optimal control system, the sub-optimal transferring time is reserved whereas the energy consumption is reduced in comparison with the no-switched scenario. In order to reflect such understanding, in the paper, two above-mentioned objectives are distributed into three sub-intervals of the transferring process. Specifically, the goal in retaining the sub-optimal transferring time and suppressing the residual vibration will be carried out by two intervals, namely acceleration and deceleration stages. Note that, in these two stages, the clutch is engaged so that the computed optimal control input can be applied; hence, the energy is consumed. On the other hands, the objective in reducing the energy consumption is shaped into the free-run interval, in which the clutch is disengaged and the payload travels by its own angular momentum generated by the previous acceleration stage; therefore, no energy consumption happens in the free-run interval. By combining three aforementioned sub-intervals, the main purpose in retaining the sub-optimal transferring time, suppressing the residual vibration, and reducing the energy consumption can be achieved.

At this point, it is recognized that in conjunction to the task in computing optimal control inputs for the actuating intervals, one must determine two switching instants t_1 and t_2 at which the clutch is disengaged and engaged respectively. One solution to this problem is to impose specific state constraints at the first switching instant t_1 so that the original switched optimal control problem can be transformed to the no-switched domain, which leads to the particular solution. On the other hands, if the system state at t_1 and t_2 are allowed to be free then the switching instants need to be optimized, which results in the general solution. Overview and relation between two types of solutions are illustrated in Fig. 10.

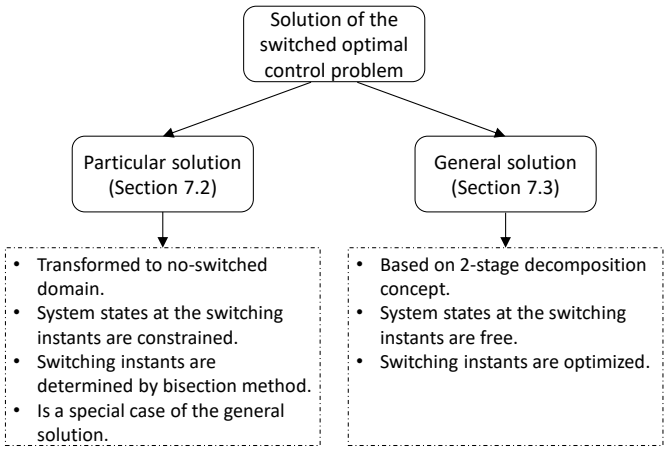

Figure 10: Two solutions of the switched optimal control problem.

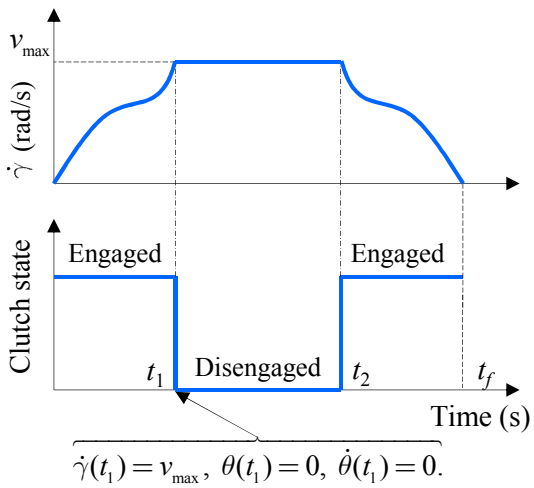

Figure 11: Particular solution: profile of the payload speed $\dot{\gamma}(t)$ with specified state constraints at the switching instant t_1.

7.2. Particular solution

The basic concept of the particular solution is to artificially impose the state constraints (24) at the switching instant t_1

$$\dot{\gamma}(t_1) = v_{\max},\ \theta(t_1) = 0,\ \dot{\theta}(t_1) = 0. \tag{24}$$

A graphical illustration of the particular solution is shown in Fig. 11. The optimal trajectory of the payload speed $\dot{\gamma}(t)$ is partitioned into three intervals: acceleration, free-run, and deceleration, with a set of state constraints at the end of acceleration stage. Note that, in the particular solution, the control input of jth interval will be denoted as $\tilde{u}_j$ $(j = 1, 3)$ whose optimums are $\tilde{u}_j^\star(t)$.

In order to further explain the choice of the constraints made in (24), recall the subsystem dynamic when the clutch is disengaged as follows

$$\begin{cases} \ddot{\gamma}(t) &= 0 \\ \ddot{\theta}(t) &= f\left(\theta(t), \dot{\theta}(t)\right). \end{cases} \tag{25}$$

According to (24) and (25), it is guaranteed that $\dot{\gamma}(t_2) = v_{\max}$, $\theta(t_2) = 0$, and $\dot{\theta}(t_2) = 0$. Therefore, when the bounded optimal control input $\tilde{u}_1^\star$ is found in the acceleration interval in order to bring the system from the initial condition to the intermediate state specified in (24), then the optimal control input $\tilde{u}_3^\star$ of the deceleration stage can be simply flipped and sign reversed. Note that, by (24), thus $\dot{\varphi}(t_1) = \dot{\gamma}(t_1) + \dot{\theta}(t_1) = v_{\max} + 0 = v_{\max}$. This means that the motor will use its maximum velocity at the end of the acceleration stage to "throw" the payload.

By pre-specifying the system states at the switching instant t_1 by (24), the optimal control problem for the switched system can be now reduced to a problem of finding the bounded optimal control input for the acceleration stage which is actually no-switched:

$$\begin{aligned} \min_{\tilde{u}_1} \quad & J_p = \phi_p\left(\boldsymbol{x}(t_1)\right) + \underbrace{\int_{t_0}^{t_1} k_1 \phi_1^2 dt}_{\text{penalty term}} \\ s.t. \quad & \dot{\boldsymbol{x}} = \boldsymbol{f}_1(\boldsymbol{x}) + \boldsymbol{b}_1(\boldsymbol{x})\tilde{u}_1 \\ & |\tilde{u}_1| \le u_{\max} \\ & \phi_p = \omega_2\left(x_2(t_1) - v_{\max}\right)^2 + \omega_3 x_3^2(t_1) + \omega_4 x_4^2(t_1) \\ & \phi_1 = v_{\max}^2 - (x_2 + x_4)^2 \\ & k_1 = \begin{cases} 0 & \text{if } \phi_1 \ge 0 \\ \mu_1 & \text{if } \phi_1 < 0 \end{cases}, \end{aligned} \tag{26}$$

where ϕ_1 regards to the state constraint due to the maximum of the servo motor speed. The weighting coefficients are similar to those used in the no-switched scenario, namely $\omega_2 = \omega_3 = \omega_4 = \mu_1 = 10^6$. The optimal control problem (26) can be solved by means of the conventional clipping-off CG method described in section 5, where the initial guess of the optimal control input is chosen as $\tilde{u}_1^0(t) = u_{\max}$, $\forall t \in \mathbb{T}_1$.

The switching instant t_1 is specified as the sub-optimal time which is successfully found such that $\exists \tilde{u}_1^\star(t_1)$: $J_p(\tilde{u}_1^\star(t_1)) < J_{tol}$ by using the bisection method in a similar manner with Algorithm 1. Note that the starting value of t_1 in the bisection iteration can be chosen as $v_{\max}/u_{\max}$. Once the optimal control input $\tilde{u}_1^\star(t)$ and the first switching instant t_1 are found, the second switching instant t_2 can be automatically determined by

$$t_2 = \frac{\gamma_d - 2\iint_0^{t_1} \tilde{u}_1^\star(t) dt}{v_{\max}} + t_1 \tag{27}$$

The computation is conducted on a personal computer with 2.5-GHz Intel Core i5-3210M and 8 GB of RAM. It takes 24.6 seconds for the particular solution to converge at the optimum. The optimal control input $\tilde{u}_1^\star(t)$ and the correspondent state trajectories are shown in Fig. 12. The resultant sub-optimal switching instant and cost functional are given as $t_1^\star = 1.88$ seconds and $J_p(\tilde{u}_1^\star(t_1^\star)) = 0.2571$ respectively. From Fig. 12, one can easily compute the profile of $\dot{\varphi}(t)$ which is then fed to the servo motor as a velocity

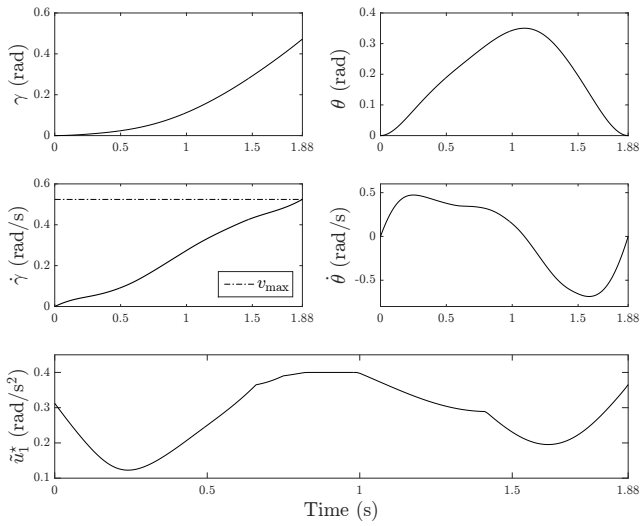

Figure 12: Control input and state trajectories at the optimum obtained by the particular solution.

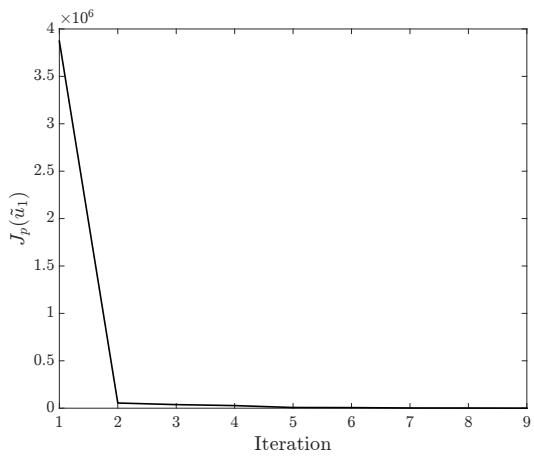

Figure 13: Cost function history over iterations of the particular solution at the resultant switching instant $t_1 = 1.88$ seconds.

command. The implementation of the switched optimal control system using the particular solution is provided in section 8. The cost functional history over iterations at $t_1 = 1.88$ seconds is shown in Fig. 13. It can be observed that the clipping-off CG method improves the cost functional of the initial guess $J_p(\tilde{u}_1^0) = 3.87 \times 10^6$ to the optimum $J_p(\tilde{u}_1^\star) = 0.2571$ in only a few iterations. The requirements of the control performance is fulfilled since $J_p(\tilde{u}_1^\star) < J_{tol}$, $(J_{tol} = 1)$. Thus, the particular solution is feasible to solve the switched optimal control problem for the rotary hook system.

Remark 5. Since the acceleration and deceleration intervals are explicitly partitioned in the particular solution hence one needs to determine the lower bound of γ_d for which the particular solution is applicable. This lower bound of is given as

$$\underline{\gamma_d} = 2 \iint_0^{t_1} \tilde{u}_1^\star(t) dt. \tag{28}$$

Based on the optimal control input $\tilde{u}_1^\star$ shown in Fig. 12, it can be numerically computed as $\underline{\gamma_d} = 49.8$ degrees. Implication of the lower bound $\underline{\gamma_d}$ is given in the section 9. Note that for any target skewing angle γ_d satisfying the condition $\gamma_d > \underline{\gamma_d}$ then the optimal solution shown in Fig. 12 can be reusable. The only parameter that needs to be recomputed is the switching instant t_2 which can be easily obtained by (27).

In summary, the total optimal control input for the whole transferring process in the case of the particular solution is denoted as $\tilde{u}^\star(t)$, in which $\tilde{u}^\star(t) = \tilde{u}_1^\star(t)$ $(t \in \mathbb{T}_1)$, $\tilde{u}^\star(t) = 0$ $(t \in \mathbb{T}_2)$, and $\tilde{u}^\star(t) = \tilde{u}_3^\star(t)$ $(t \in \mathbb{T}_3)$.

7.3. General solution

As noted in Remark 5, the particular solution is only applicable when the reference skewing angle is sufficiently large, namely $\gamma_d > \underline{\gamma_d} = 49.8$ degrees. Therefore, in this section, an attempt is made to derive a general solution to overcome the above-mentioned limitation of the particular solution. In order to do so, the state constraints (24) at the switching instant t_1 will be nullified and the switching instants will be optimized. Note that, to draw a distinction with the particular solution, the control input of jth interval in the general solution will be denoted as u_j $(j = 1, 3)$ whose optimums are $u_j^\star(t)$.

Now, the switching instants optimization problem arises together with the common problem in computing the bounded optimal control input with state constraints. As usual, to eliminate the state constraint, a step penalty function is employed. Therefore, the optimal control problem with a fixed terminal time for the switched rotary hook system is stated as follows

$$\min_{u_1, u_3, t_1, t_2} J_g = \phi\left(\boldsymbol{x}(t_f)\right) + \underbrace{\int_{t_0}^{t_f} L(\boldsymbol{x}) dt}_{\text{Penalty term}} \tag{29}$$

$s.t.$

$$\dot{\boldsymbol{x}} = \begin{cases} \boldsymbol{f}_k(\boldsymbol{x}) + \boldsymbol{b}_k(\boldsymbol{x}) u_k, & \text{if } t \in \mathbb{T}_k \ (k = 1, 3). \\ \boldsymbol{f}_k(\boldsymbol{x}), & \text{if } t \in \mathbb{T}_k \ (k = 2). \end{cases}$$

$$\phi\left(\boldsymbol{x}(t_f)\right) = \omega_1 \left[x_1(t_f) - \gamma_d\right]^2 + \omega_2 x_2^2(t_f) + \omega_3 x_3^2(t_f) + \omega_4 x_4^2(t_f)$$

$$L(\boldsymbol{x}) = k_1 \phi_1^2$$

$$|u_1| \leq u_{\max}, \ |u_3| \leq u_{\max}$$

$$\phi_1 = v_{\max}^2 - (x_2 + x_4)^2$$

$$k_1 = \begin{cases} 0 & \text{if } \phi_1 \geq 0 \\ \mu_1 & \text{if } \phi_1 < 0, \end{cases}$$

In (29), the weighting coefficients are $\omega_1 = \ldots = \omega_4 = \mu_1 = 10^6$. As can be seen from (29), one needs to compute not only bounded optimal control inputs with state constraints but also the optimal switching instants t_1 and t_2. For this reason, a general solution to (29) will be formulated based on a concept of two-stage decomposition, namely optimal control seeking stage (stage (a)) and optimal switching instants seeking stage (stage (b)) [37, 38]. In

Algorithm 2: General solution of the switched optimal control problem for the rotary hook system by means of the two-stage decomposition concept.

Step 1 Set the incremental index $i = 0$. Choose an initial switching instants vector $\boldsymbol{t}^0 = \begin{bmatrix} t_1^0 & t_2^0 \end{bmatrix}^\top$.

Step 2 -Stage (a)- Solving the optimal control problem for the system with respect to the switching instant vector $\boldsymbol{t}^i$. Obtain $J_g(\boldsymbol{t}^i)$. Go to step 3.

Step 3 -Stage (b)- By using $J_g(\boldsymbol{t}^i)$, update the switching instants vector by solving the following constrained optimization problem

$$\min_{0<t_1<t_2<t_f} \quad J_g(t_1, t_2) \tag{30}$$

By means of a nonlinear programming method, the switching instants can be updated. Denote the updated value of $\boldsymbol{t}^i$ as $\boldsymbol{t}^{i+1}$. If the nonlinear programmer converges to a local optimum then terminate. Otherwise, let $i = i + 1$ and repeat step 2.

stage (a), switching instants are assumed to be fixed thus the main task is to compute the bounded optimal control input with state constraints. By means of this stage, one can obtain the cost functional J_g corresponding to the current switching instants. In stage (b), the switching instants are updated by employing the cost functional J_g obtained in stage (a) through any constrained nonlinear programming technique. The conceptual procedure is illustrated in Algorithm 2. Details of both stage (a) and stage (b) will be briefly explained as follows. To solve stage (b), the interior point technique provided by the *fmincon* routine of MATLAB will be straightforwardly employed. However, in order to solve stage (a) in which the control input constraints present, an extension of the clipping-off CG method needs to be further developed to suit for the nonlinear switched system context. In the first step, necessary conditions for a control input to be optimal in respect of a switched system with known switching instants must be formulated. In [37], such necessary conditions are derived only for a switched system without control input constraints. In this paper, it will be extended for a switched system with bounded control inputs by means of Lemma 1 (Appendix A). Based on Lemma 1, an extended clipping-off CG scheme is proposed in Algorithm 3 of the same appendix. By using Algorithm 3, the switched optimal control problem, subject to the fixed switching instants posed in stage (a), can be solved. The convergence analysis of Algorithm 3 is given in Appendix B.

In the general solution, the sub-optimal transferring time can also be achieved by extending the fixed terminal time switched optimal control problem (29) by using the bisection method in a similar manner to Algorithm 1. However, one cannot expect a faster transferring time compared with the no-switched control system (see Remark 4). Therefore, it is rational to utilize the successive sub-optimal transferring time found in the no-switched optimal control solution shown in the section 6 as the target terminal time for the general solution.

Now, it is ready to solve the switched optimal control problem (29) by the general solution. As the above discussions, the target terminal time t_f for the switched control system is given by the successive sub-optimal transferring time found in the no-switched scenario, e.g. $t_f = 5.06$ seconds for the medium transferring case ($\gamma_d = 90$ degrees) and $t_f = 7.97$ seconds in the long transferring case ($\gamma_d = 180$ degrees). To start the procedure, initial guesses of the necessary terms are given as

$$u_1^0(t) = u_{\max}, \forall t \in \mathbb{T}_1 \text{ and } u_3^0(t) = -u_{\max}, \forall t \in \mathbb{T}_3.$$
$$\boldsymbol{t}^0 = \begin{bmatrix} t_1^0, & t_2^0 \end{bmatrix}^\top = [t_f/3, \ 2t_f/3]^\top.$$

Let $\boldsymbol{u}^0(t) = \begin{bmatrix} u_1^0(t), & u_2^0(t) \end{bmatrix}^\top$ and denote the resultant optimal control input of the general solution as $\boldsymbol{u}^\star(t) = [u_1^\star(t), \ u_2^\star(t)]^\top$. In the medium transferring case, it takes more than 40 minutes (in 42 iterations) for the general solution given by Algorithm 2 to converge. The optimized switching instants are achieved as $\boldsymbol{t}^\star = [t_1^\star, \ t_2^\star]^\top = [1.72, \ 3.48]^\top$ seconds. The cost functional is improved from initial value $J_g\left(\boldsymbol{u}^0(t), \boldsymbol{t}^0\right) = 7.2 \times 10^4$ to the local minimum $J_g\left(\boldsymbol{u}^\star(t), \boldsymbol{t}^\star\right) = 289.2$. Note that the converged solution cannot satisfy the required performance since $J_g\left(\boldsymbol{u}^\star(t), \boldsymbol{t}^\star\right) > J_{tol} = 1$. Similar situations are encountered in the short and long transferring cases.

Following discussions are provided to clarify the unsatisfactory results obtained by the general solution. Firstly, the undesirable performance index can be explained by the fact that both clipping-off CG method in the stage (a) and nonlinear programming in the stage (b) only result in local optimum, thus the converged cost functional $J_g\left(\boldsymbol{u}^\star(t), \boldsymbol{t}^\star\right)$ is conservative. Secondly, the long computational time is due to the switching instants optimization (stage (b)), which significantly increase the computational burden since the stage (a) must be restarted all over again in each iteration when the switching instants are updated.

From the above unsatisfactory result of the proposed general solution in the paper, it opens some possibilities in improving the performance of the general solution to extend the feasibility of energy saving by using the switched optimal control system in practice.

8. Experimental results of the switched control system

Effectiveness of the switched optimal control system is now verified on the experimental apparatus. Two cases, namely medium transferring ($\gamma_d = 90$ degrees) and long tranferring ($\gamma_d = 180$ degrees), will be validated in the experiments. By means of the particular solution, computational results are given in the Table 3. It is worth

mentioning that during the free-run interval, namely when clutch is disengaged, the motor is actually turned off.

Table 3: Computational results of the particular solution for the switched control system.

Case	SW instants	Trans. time[a]	Compt. time[b]
$\gamma_d = 90°$	$t_1 = 1.88$ s $t_2 = 3.07$ s	4.95 s	24.6 s
$\gamma_d = 180°$	$t_1 = 1.88$ s $t_2 = 6.06$ s	7.94 s	24.6 s

[a] Transferring time, [b] Computational time.

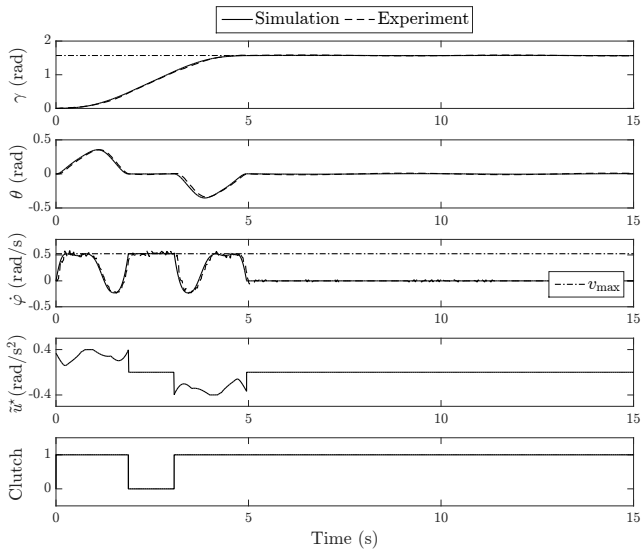

Figure 14: Simulation and experimental results of the switched control system in the medium transferring case. In the clutch graph, "1" denotes the engagement whereas "0" represents the disengagement.

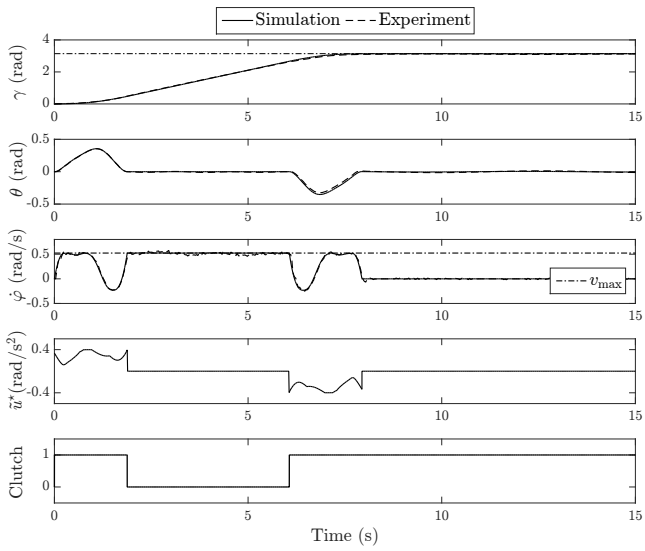

Figure 15: Simulation and experimental results of the switched control system in the long transferring case. In the clutch graph, "1" denotes the engagement whereas "0" represents the disengagement.

Experimental results of the switched control system are shown in Fig. 14 and Fig. 15. Note that in each figure, the clutch state are presented in which "1" and "0" represent the engaging and disengaging configuration respectively. It is recognized that the experimental results in all cases are in total agreement with the simulation results. The payload precisely arrives at the desired skew angle whilst the skew oscillation is nearly suppressed at the end of the motion. Therefore, performance of the switched optimal control system employing the particular solution is confirmed.

9. Comparison between switched and no-switched control systems

In this section, two scenarios namely switched and no-switched control systems will be compared in terms of the transferring time and overall energy consumption. The actual energy consumption of the servo motor will be employed to quantitatively access the energy usage of each control system. Specifically, the power utilized by the servo motor is given by $P(t) = \tau_m \dot{\varphi}_m(t)$, where $\tau_m(t)$ is the motor torque. Furthermore, $\tau_m(t)$ is related to the actual motor current $i_m(t)$ by a torque constant $k_t = 0.0152$ (Nm/A), therefore $P(t) = k_t i_m(t)\dot{\varphi}_m(t)$. Energy consumption of the servo motor is just an integration of the power over the operating time, namely

$$E = k_t \int_0^{t_f} i_m(t)\dot{\varphi}_m(t)dt\Big|_{i_m(t)\dot{\varphi}_m(t)\geq 0} \tag{31}$$

Note that the regions where $P(t) = k_t i_m(t)\dot{\varphi}_m(t) < 0$ are the regenerative braking intervals. In these intervals, the power flows back to the system and a dumped resistor is used to dissipate this power into heat. Therefore, the motor does not use energy hence the regenerative braking periods are excluded in (31). In the experiments, the actual current $i_m(t)$ and the velocity $\dot{\varphi}_m(t)$ are measured for each switched/no-switched control system. Therefore, the energy consumption E for each controller can be numerically computed by (31).

In the experimental studies, the transferring time t_c is defined as the time instant which satisfies two conditions: a) the response curve of the payload's absolute angle $\gamma(t)$ reaches and maintains within a range ζ% of the final target angle γ_d, b) the vibration nearly vanishes; that is to say

$$t_c = \min\{t : \forall \Delta_t > 0,\ (33) \text{ and } (34) \text{ are fufilled}\}. \tag{32}$$

The conditions (33) and (34) are defined as follows

$$(1 - \zeta \mathrm{sgn}(\gamma_d))\gamma_d < \gamma(t + \Delta_t) < (1 + \zeta \mathrm{sgn}(\gamma_d))\gamma_d, \tag{33}$$

$$|\theta(t + \Delta_t)| < |\theta(t)| \leq \theta_{tol},\ . \tag{34}$$

Here, it is chosen that $\zeta = 1\%$, $\theta_{tol} = 1$ degree. Note that "sgn" represents a usual signum function.

Table 4: Transferring time and energy consumption of switched and no-switched control systems.

		Trans. time t_c	Energy Q	Comp. time
$\gamma_d = 90°$	**SW**[a]	**4.83 s**	**3.04 J**	**24.6 s**
	NSW[b]	4.89 s	4.08 J	61.7 s
	IS[c]	5.78 s	3.82 J	4.2 s
	ISMC[d]	6.91 s	3.37 J	—
$\gamma_d = 180°$	**SW**	**7.91 s**	**3.11 J**	**24.6 s**
	NSW	7.93 s	8.12 J	85.2 s
	IS	11.74 s	7.10 J	5.4 s
	ISMC	10.61 s	7.18 J	—

[a] Switched optimal control, [b] No-switched optimal control, [c] Input shaping control, [d] Integral sliding mode control. — Real Time.

A comparative results between the switched and no-switched optimal controls in the medium and long transferring cases are shown in Fig. 16 and Fig. 17 respectively. Their transferring times and energy consumptions are summarized in Table 4. According to Table 4, the switched optimal control system is able to retain the sub-optimal transferring times provided by the no-switched optimal solution. For instance, in the long transferring case ($\gamma_d = 180°$), the sub-optimal transferring time of the switched and no-switched schemes are 7.91 seconds and 7.93 seconds respectively. Furthermore, when the switched optimal control system is employed, the energy consumption is respectively reduced up to 25.49% and 61.70% in the medium and long transferring cases compared with the no-switched counterpart. Therefore, our purpose in *saving energy* for the rotary hook system is achieved *without any influence to the total transferring time* by employing the switched optimal control approach.

It should be noted that the particular solution is only applicable when the target skew angle γ_d satisfies the condition that $\gamma_d > \underline{\gamma_d}$, namely γ_d needs to be sufficiently large. Therefore, in practice, no-switched and switched optimal control systems will be used in a complementary manner. For the short transferring cases, namely $\gamma_d < \underline{\gamma_d} = 49.8$ degrees, no-switched scenario is put into operation. Otherwise, the switched optimal control system will take place to reduce the energy consumption. Such operation is rational since in the short transferring contexts, the amount of energy saving is not significant. Furthermore, in these cases, switching within a small period of time will possibly causes damage to the clutch.

In order to clarify the effectiveness of the proposed switched optimal control scheme in reducing both transferring time and energy consumption, feed-forward Input Shaping (IS) control and Integral Sliding Mode Control (ISMC) are intentionally applied in the context where the clutch is entirely engaged during the motion. The ISMC is used in an exact manner to [19] whereas in the input shaping scheme, the fastest ZV input shaper [39] is employed.

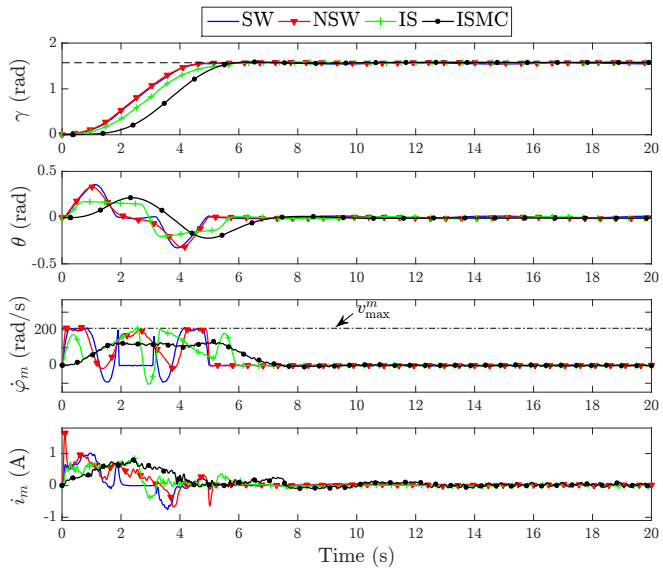

Figure 16: Comparative experimental results of the switched optimal, no-switched optimal, input shaping, and integral sliding mode controllers ($\gamma_d = 90°$).

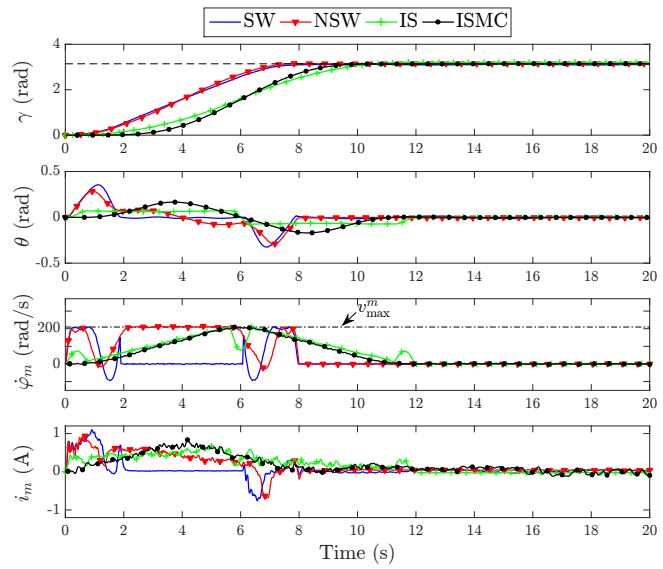

Figure 17: Comparative experimental results of the switched optimal, no-switched optimal, input shaping, and integral sliding mode controllers ($\gamma_d = 180°$).

Since the damping ratio of the rotary hook system is not considered, magnitude A_i and time location τ_i ($i = 1, 2$) of each impulse in the ZV input shaper are given as follows

$$\begin{bmatrix} A_i \\ \tau_i \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0 & \dfrac{\pi}{\omega_n} \end{bmatrix}, \tag{35}$$

where ω_n denotes the angular natural frequency of the system. By linearizing the nonlinear system (6) around the equilibrium point, it can be found that $\omega_n = \sqrt{mgR^2/(l \times I_H)}$ and by using the values in Table 1, it can be obtained that $\omega_n = 4.175$ (rad/s). Therefore, the ZV input shaper in (35) is now specified. In order to en-

sure a smooth motion, following second order trajectory is planned as the original command for the input shaping control

$$r_{IS}(t) = \begin{cases} \dfrac{4\gamma_d}{\rho^2}\dfrac{t^2}{2} & \text{if } 0 \le t \le \rho/2 \\ \dfrac{\gamma_d}{\rho^2}\left(-2t^2 + 4\rho t - \rho^2\right) & \text{if } \rho/2 \le t \le \rho \\ \gamma_d & \text{if } t \ge \rho. \end{cases} \quad (36)$$

In (36), ρ regards to the terminal time at which the reference trajectory $r_{IS}(t)$ reaches the target skew angle γ_d. By gradually reducing ρ in a similar manner to the bisection method shown in Algorithm 1, the smallest transferring time (in case of the input shaping control) subjected to the state constraint $|\dot{\varphi}(t)| \le v_{\max}$ and the control input constraint $|u(t)| \le u_{\max}$ can be achieved. Note that the starting value of ρ in the bisection iteration can be chosen as $2\gamma_d/v_{\max}$.

The experimental results of the input shaping control and ISMC are shown in Fig. 16 - Fig. 17 for the medium and long transferring case respectively. The transferring time, energy consumption, and computational time of all control methods are aggregated in Table 4. It can be seen that by using the switched optimal control system, when $\gamma_d = 90°$, the transferring time and energy consumption are respectively reduced up to 16.4% and 20.4% compared to IS control, whereas those are 30.1% and 9.8% in the case of ISMC. Similarly, when $\gamma_d = 180°$, the switched optimal control system saves 32.6% and 56.2% in the transferring time and energy consumption compared to IS control, whilst 25.4% and 56.7% are obtained in the case of ISMC. Note that the energy reduction percentage does not depend on the torque constant k_t. In summary, it is concluded that the proposed switched optimal controller performs a faster transferring process with a smaller energy consumption.

It should be remarked that the optimal controllers proposed in this paper are actually feed-forward schemes. Therefore, if the working environment is surrounded by various sources of disturbances such as wind, collision, etc., the proposed optimal controllers can serve as a sub-optimal-time reference trajectory planner for combining with a feedback controller in a 2-DOF control system to enhance overall performance. Furthermore, in the case that system parameters do not significantly vary, namely the parametric uncertainties are small, the proposed scheme is proved to be an effective method to reduce both transferring time and energy consumption for the payload's skew rotation system.

10. Conclusions and future work

In this paper, nonlinear dynamic of a hybrid rotary hook system is formulated in theory and validated in experiments. Two types of optimal control systems, namely switched and no-switched, are proposed to drive the payload to a desirable skew orientation without any residual vibration. The novelty of the switched optimal controller lies in the fact that it is able to reduce energy consumption without trading-off the sub-optimal transferring time. The key point is to use an electro-mechanical clutch to generate a free motion of the payload imposing upon its own inertial movement. In the paper, to resolve the switched optimal control problem, both general and particular solutions are proposed in which the particular solution is shown to be advantageous over the general solution. The basic idea of the particular solution is to transform the original switched optimal control problem to the no-switched domain by imposing the state constraints at the switching instants. In comparison to other candidates, namely input shaping and integral siding mode controllers, the proposed switched optimal control scheme shows its advantages in reducing both transferring time and energy consumption. Effectiveness of the optimal control systems are demonstrated on an experimental apparatus and their validities have been shown. At present, the interference between the skewing rotation and the swinging dynamic is not considered. In our future works, the skewing rotation dynamic will be integrated into the swinging dynamic of the crane systems to design a comprehensive automatic controller, which can suppress not only the swing angles but also residual skew oscillation of the payload simultaneously during the transferring process.

Appendix A. Details of the general solution

This appendix will provide details of the extended CG scheme, which is used by the general solution described in section 7.3, to solve the switched optimal control seeking stage (stage (a) of Algorithm 2).

In the first step, necessary conditions of optimality will be derived by means of following lemma.

Lemma 1. *Consider the optimal control problem (29). Assuming that the switching instants are now fixed. The necessary conditions for optimality are given as*

$$\dot{\boldsymbol{x}} = \begin{cases} \boldsymbol{f}_k(\boldsymbol{x}) + \boldsymbol{b}_k(\boldsymbol{x})u_k & (k = 1,3) \\ \boldsymbol{f}_k(\boldsymbol{x}) & (k = 2) \end{cases}, \ \textit{if } t \in \mathbb{T}_k. \quad (A.1)$$

$$\dot{\boldsymbol{\lambda}}(t) = -\left(\frac{\partial H_k}{\partial \boldsymbol{x}}\right)^\top \quad (k = 1,2,3), \ \textit{where} \quad (A.2)$$

$$H_k \triangleq \begin{cases} L(\boldsymbol{x}) + \boldsymbol{\lambda}^\top\left(\boldsymbol{f}_k(\boldsymbol{x}) + \boldsymbol{b}_k(\boldsymbol{x})u_k\right) & (k = 1,3) \\ L(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{f}_k(\boldsymbol{x}) & (k = 2) \end{cases},$$

$$\boldsymbol{\lambda}(t_f) = \left(\frac{\partial \phi\left(\boldsymbol{x}(t_f)\right)}{\partial \boldsymbol{x}(t_f)}\right)^\top, \quad (A.3)$$

$$\boldsymbol{\lambda}(t_k^+) = \boldsymbol{\lambda}(t_k^-) \quad (k = 1,2), \quad (A.4)$$

$$u_k = \begin{cases} u_{\max} & \text{if } \dfrac{\partial H_k}{\partial u_k} < 0 \\ -u_{\max} \le u_k \le u_{\max} & \text{if } \dfrac{\partial H_k}{\partial u_k} = 0 \\ -u_{\max} & \text{if } \dfrac{\partial H_k}{\partial u_k} > 0 \end{cases}, \ (k = 1, 3). \tag{A.5}$$

The proof of the Lemma 1 is conducted in a similar manner with [40, p.73] therefore it is omitted here. Note that in the Lemma 1, (A.1) represents the switched system dynamic in each time interval. Conditions of the costate vector $\boldsymbol{\lambda}(t)$ are given in (A.2)-(A.4). The bounded optimal control input of each interval must satisfy (A.5).

Based on Lemma 1, following extended clipping-off CG algorithm is proposed to solve the switched optimal control problem subject to the fixed switching instants posed in the stage (a). From (A.5), it is motivated to define 2 sub-interval point sets, "unsaturated-control sub-interval points" $\mathbb{U}_j^i$ and "saturated-control sub-interval points" $\mathbb{S}_j^i$ of jth interval ($j = 1, 3$) at ith iteration as follows.

$$\mathbb{S}_j^i = \left\{ t : \left| u_j^i(t) \right| = u_{\max} \text{ and } \operatorname{sgn}(u_j^i(t)) = -\operatorname{sgn}(g_j^i(t)) \right\}, \tag{A.6}$$

$$\mathbb{U}_j^i = \mathbb{T}_j - \mathbb{S}_j^i, \ t \in \mathbb{T}_j, \ j = 1, 3, \tag{A.7}$$

where $g_j^i(t) \triangleq \partial H_j^i / \partial u_j^i$ and "sgn" represents a usual signum function. In Algorithm 3, the notation $(.)_j^i(t)$ denotes the value of the function (.) at ith iteration and in jth interval.

Algorithm 3: Extended clipping-off CG algorithm for solving the bounded optimal control input in respect of fixed switching instants and fixed terminal time.

Step 1 At every time step $t \in \mathbb{T}_1 \cup \mathbb{T}_3$, giving an initial guess of the optimal control input $\boldsymbol{u}^0(t) = \left[u_1^0(t), \ u_3^0(t) \right]^\top \in \Omega$. Solving the state equation (A.1) forward and costate equation (A.2) backward in time. Compute $\boldsymbol{g}^0(t) = \left[g_1^0(t), \ g_3^0(t) \right]^\top = \left[\dfrac{\partial H_1^0}{\partial u_1^0}, \ \dfrac{\partial H_3^0}{\partial u_3^0} \right]^\top$ by employing the continuity condition in (A.4). Let the initial search direction as $\boldsymbol{p}^0(t) \triangleq \left[p_1^0(t), \ p_3^0(t) \right]^\top = -\boldsymbol{g}^0(t)$.

Step 2 Choose a scalar step-size α^0 such that $J_g\left(\left\{ \boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0 \boldsymbol{p}^0 \right\}_{\boldsymbol{s}} \right)$ is minimized by means of any one dimensional linesearch, where the value of $\boldsymbol{\varepsilon}^0$ at the time step t is given as $\boldsymbol{\varepsilon}^0(t) = \left[\varepsilon_1^0(t), \ \varepsilon_3^0(t) \right]^\top$ and

$$\varepsilon_j^0(t) = \begin{cases} 0 & \text{if } t \in \mathbb{S}_j^0 \\ \alpha^0 > 0 & \text{if } t \in \mathbb{U}_j^0 \end{cases}, \ (j = 1, 3). \tag{A.8}$$

For clarification, $\left\{ \boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0 \boldsymbol{p}^0 \right\}_{\boldsymbol{s}}$ at t is given by

$$\left\{ \boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0 \boldsymbol{p}^0 \right\}_{\boldsymbol{s}} = \begin{bmatrix} \left\{ u_1^0(t) + \varepsilon_1^0(t) p_1^0(t) \right\}_s \\ \left\{ u_3^0(t) + \varepsilon_3^0(t) p_3^0(t) \right\}_s \end{bmatrix}, \tag{A.9}$$

and

$$\{(.)\}_s = \begin{cases} (.) & \text{if } |(.)| \le u_{\max} \\ u_{\max} & \text{if } (.) > u_{\max} \\ -u_{\max} & \text{if } (.) < -u_{\max}. \end{cases} \tag{A.10}$$

Let $\boldsymbol{u}^1 = \left\{ \boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0 \boldsymbol{p}^0 \right\}_{\boldsymbol{s}}$ and $i = 1$.

Step 3 At each t, determine the next search direction $\boldsymbol{p}^i(t) = \left[p_1^i(t), \ p_3^i(t) \right]^\top$ as follows

$$p_j^i(t) = \begin{cases} -g_j^i(t) + \beta_j^i p_j^{i-1}(t) & \text{if } t \in \mathbb{U}_j^i, \\ -g_j^i(t) & \text{if } t \in \mathbb{S}_j^i, \end{cases} \tag{A.11}$$

where $j = 1, 3$ and

$$\beta_j^i = \begin{cases} (\beta_0^i)_j & \text{if } (\beta_0^i)_j < (\beta_m^i)_j \\ \gamma^i (\beta_m^i)_j & \text{if } (\beta_0^i)_j \ge (\beta_m^i)_j, \ 0 < \gamma^i < 1 \\ 0 & \text{if } (I_2^i)_j = 0 \text{ or } (I_3^i)_j = 0, \end{cases} \tag{A.12}$$

with $(\beta_0^i)_j \triangleq \dfrac{(I_1^i)_j}{(I_2^i)_j}$, $(I_2^i)_j \neq 0$; $(\beta_m^i)_j \triangleq \dfrac{(I_1^i)_j}{(I_3^i)_j}$, $(I_3^i)_j \neq 0$, and

$$\begin{aligned} (I_1^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^i(t) g_j^i(t) dt \\ (I_2^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^{i-1}(t) g_j^{i-1}(t) dt \\ (I_3^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^i(t) p_j^{i-1}(t) dt. \end{aligned} \tag{A.13}$$

Step 4 Choose a step-size α^i to minimize $J_g\left(\left\{ \boldsymbol{u}^i + \boldsymbol{\varepsilon}^i \boldsymbol{p}^i \right\}_{\boldsymbol{s}} \right)$ in the similar manner with step 2, then let $\boldsymbol{u}^{i+1} = \left\{ \boldsymbol{u}^i + \boldsymbol{\varepsilon}^i \boldsymbol{p}^i \right\}_{\boldsymbol{s}}$ and $i \leftarrow i + 1$. Go to step 5.

Step 5 Repeat step 3 and step 4 until $\left| J_g(\boldsymbol{u}^{i+1}) - J_g(\boldsymbol{u}^i) \right| < \delta \left| J_g(\boldsymbol{u}^{i+1}) \right|$ or $J_g(\boldsymbol{u}^{i+1}) < J_{tol}$ where $\delta > 0$ regards to tolerance at which the algorithm is said to be converged. Here, the tolerance δ is chosen as $\delta = 10^{-6}$.

It is noted that, in the step 2 and 4 of Algorithm 3, the minimizer α^i of $J\left(\left\{ \boldsymbol{u}^i + \boldsymbol{\varepsilon}^i \boldsymbol{p}^i \right\}_{\boldsymbol{s}} \right)$ can be obtained by any one-dimensional linesearch method. In this paper, the golden section method [41] is employed. Specifically, a high-low-high geometry pattern of the cost functional J_g with respect to α^i should be found beforehand to determine the feasible searching interval.

The convergence analysis of the Algorithm 3 is shown in the Appendix B.

Appendix B. Convergence analysis of Algorithm 3

In this appendix, inspired from [32], convergence analysis will be carried out for Algorithm 3. By means of the

one-dimensional linesearch in the Step 4, variation of the control input u_j^i in jth interval at ith iteration is given as follows

$$\delta u_j^i = \begin{cases} \alpha^i p_j^i(t) & \text{if } t \in \mathbb{U}_j^i \\ 0 & \text{if } t \in \mathbb{S}_j^i \cup \mathbb{T}_2 \end{cases}, (j = 1, 3). \tag{B.1}$$

Due to (B.1) and (A.2)-(A.4), variation of the cost functional J_g^i at ith iteration is described by

$$\delta J_g^i = \int_{t_0}^{t_f} \frac{\partial H}{\partial u} \delta u \bigg|^i dt = \sum_{j=1,3} \int_{t_{j-1}}^{t_j} \frac{\partial H}{\partial u} \delta u \bigg|_j^i dt \tag{B.2}$$

$$= \alpha^i \sum_{j=1,3} \int_{\mathbb{U}_j^i} g_j^i(t) p_j^i(t) dt. \tag{B.3}$$

It is noted that $t_3 \triangleq t_f$. Substituting (A.11) into (B.3) results in

$$\delta J_g^i = \alpha^i \sum_{j=1,3} \int_{\mathbb{U}_j^i} \left[-g_j^i(t) g_j^i(t) + \beta_j^i g_j^i(t) p_j^{i-1}(t) \right] dt \tag{B.4}$$

$$= \alpha^i \sum_{j=1,3} \left[-(I_1^i)_j + \beta_j^i (I_3^i)_j \right] \tag{B.5}$$

According to (A.12)-(A.13), $-(I_1^i)_j + \beta_j^i (I_3^i)_j < 0$ can be always guaranteed and by $\alpha^i > 0$ thus $\delta J_g^i < 0$. Therefore, it can be obtained that $J_g(\boldsymbol{u}^{i+1}) < J_g(\boldsymbol{u}^i)$, $i = 0, 1, 2, \ldots$ From this, the Algorithm 3 will converge to a local minimum $J_g(\boldsymbol{u}^\star)$. When the local minimum $J_g(\boldsymbol{u}^\star)$ is reached as $i \to \infty$, (B.6) holds

$$g_j^i(t) = g_j^{i-1}(t) \text{ and } p_j^i(t) = p_j^{i-1}(t), \ (j = 1, 3). \tag{B.6}$$

Furthermore, due to $\delta J_g^i = 0$ hence from (B.3) and (B.4), it results in

$$\begin{cases} \sum_{j=1,3} \int_{\mathbb{U}_j^i} g_j^i(t) p_j^i(t) dt = 0 \\ \sum_{j=1,3} \int_{\mathbb{U}_j^i} \left[-g_j^i(t) g_j^i(t) + \beta_j^i g_j^i(t) p_j^{i-1}(t) \right] dt = 0. \end{cases} \tag{B.7}$$

In (B.7), by employing (B.6) and substituting the upper equation into the lower equation, it can be reduced to

$$\sum_{j=1,3} \int_{\mathbb{U}_j^i} g_j^i(t) g_j^i(t) dt = 0, \tag{B.8}$$

which results in

$$g_j^i(t) \equiv 0 \text{ for every } t \text{ in } \mathbb{U}_j^i, \ (j = 1, 3). \tag{B.9}$$

By the constructions of "saturated-control sub-interval points" as in (A.6) and the result in (B.9), the necessary conditions of the optimal control input for the switched system in the stage (a) (as found in (A.5)) are now fulfilled. This completes the proof.

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